Supervised Machine Learning Classification: Enterprise Customer Churn Prediction & Retention Intelligence

Executive Analytical Report | Audience: **Chief Data Officer (CDO) & Head of Analytics** | Cohort: N = 2,500 Enterprise Accounts (12 Features) | Validation: 5-Fold Stratified CV & Holdout Test (n = 500)

RECOMMENDED CHAMPION	TEST ACCURACY / F1	ROC-AUC / PR-AUC	BASELINE CHURN RATE	NET RETENTION ROI
Gradient Boosting (GBM)	88.8% / 0.754	0.887 / 0.841	26.4% (660 / 2500)	$157,896 (Test Set)

1. Main Objective of the Analysis & Executive Key Insights (Rubric 1.1 & 1.3)

Main Objective of the Analysis: The primary objective of this analysis is to design, train, compare, and validate a **Supervised Machine Learning Classification** framework that accurately predicts whether an enterprise subscription account will **churn (cancel service, class = 1)** or **remain retained (renew service, class = 0)** within the upcoming 90-day billing cycle. Specifically, the classification model is intended to achieve two core business outcomes for the Chief Data Officer (CDO) and Customer Success leadership: (1) generate well-calibrated individual churn probability scores to prioritize high-risk enterprise accounts for proactive retention interventions before cancellation occurs, and (2) provide transparent, interpretable feature attributions that reveal the root operational and contractual drivers of customer attrition, enabling data-driven pricing and SLA reforms.

Key Insights & Strategic Findings for Stakeholders:

• **Insight 1 — Contractual Vulnerability Regime:** Accounts on *Month-to-Month* contracts exhibit a **41.6% churn rate** compared to just **10.2%** on *Two-Year* contracts. Standardized Logistic Regression confirms that transitioning a client to a Two-Year agreement reduces churn odds by **89.3% (Odds Ratio = 0.11)**.

• **Insight 2 — Behavioral Early-Warning Signal:** A 30-day product usage drop (*usage_drop_pct > 12%*) is the #1 non-linear predictor in our Champion Gradient Boosting model (Permutation Importance = 0.159), increasing churn odds by **2.63x per standard deviation**.

• **Insight 3 — Operational Service Friction Multiplier:** Accounts experiencing **4+ support tickets in 90 days** combined with an **SLA breach** or resolution time exceeding 24 hours suffer a 3.4x spike in attrition probability, proving that technical support latency directly erodes recurring revenue.

• **Insight 4 — Financial Impact of Cost-Sensitive Thresholding:** By shifting the classification probability threshold from the default $\tau = 0.50$ to a cost-optimized threshold of $\tau^* = 0.25–0.30$ (accounting for a $4,200 average contract loss per false negative vs. $180 outreach cost per false positive), our recommended **Gradient Boosting Classifier** captures **81.8% of all true churners** and protects an estimated **$157,896 in net recurring revenue** across the 500-account holdout test cohort alone (~$789,000 annualized across the 2,500-account portfolio).

2. Data Description & Feature Engineering Pipeline (Rubric 1.2)

Dataset Summary & Cohort Structure: The analysis utilizes the *Enterprise Subscription & Telecom Customer Churn Dataset* comprising **N = 2,500 customer records** and **12 explanatory features** (10 numeric/binary operational metrics and 2 nominal categorical variables) paired with a binary target outcome (*churn_status*: **0 = Retained [1,840 accounts, 73.6%]**; **1 = Churned [660 accounts, 26.4%]**). Retained customers average **28.2 months** of tenure and **$114.37** in monthly charges, whereas churned customers average **21.1 months** of tenure and **$125.16** in monthly charges.

How Features Are Used in Building, Training, and Interpreting the Classification Models: To prevent data leakage, the dataset was partitioned via **Stratified Train/Test Splitting** into an **80% Training Set (n = 2,000)** used for 5-Fold Stratified Cross-Validation and hyperparameter tuning, and a **20% Holdout Test Set (n = 500)** strictly isolated for final out-of-sample evaluation. Inside a unified *Scikit-Learn ColumnTransformer Pipeline*, continuous features were standardized via *StandardScaler (z-score normalization)* so scale-sensitive classifiers (Logistic Regression, RBF-SVM, k-NN) treat all dimensions equitably, while nominal features were encoded via *OneHotEncoder*. In model interpretation, standardized regression coefficients provide exact directional **Odds Ratios**, while **Permutation Feature Importance** and local **SHAP-style attributions** quantify how non-linear feature interactions drive tree-ensemble predictions.

Feature Variable	Data Type & Scale	Summary / Distribution	Role in Building, Training & Stakeholder Interpretation
usage_drop_pct	Numeric Continuous (%)	Range: -40% to +85%	**#1 Predictor (GBM Imp: 0.159, OR: 2.63x):** Captures 30-day behavioral disengagement prior to cancellation.
contract_type	Categorical (One-Hot)	M2M (48%), 1Yr (30%), 2Yr (22%)	**#2 Predictor (GBM Imp: 0.121):** Structural lock-in; Two-Year contracts reduce churn odds by 89.3% (OR = 0.11).

tenure_months	Numeric Continuous	1 to 72 Months (Avg: 26.3)	**Loyalty Anchor (OR: 0.66x):** Interacts non-linearly with onboarding status during the critical first 12 months.
support_tickets_90d & **avg_resolution_hours**	Numeric Count & Continuous	0–14 Tickets | 1.5–72.0 Hrs	**Service Friction Indicators:** Measure technical friction; >4 tickets coupled with >24h resolution triggers churn spikes.
sla_breach_flag & **nps_score**	Binary (0/1) & Ordinal (0–10)	SLA Breach: 18% | Avg NPS: 6.1	**Quality & Sentiment Signals:** SLA breaches raise churn odds by 42% (OR = 1.42x), especially when NPS ≤ 4.
monthly_charges_usd & **total_contract_value**	Numeric Currency ($)	$29–$249/mo | $30–$17.5k Total	**Price Sensitivity & Revenue Weighting:** Quantifies price elasticity and weights retention ROI prioritization.
active_modules_count, **onboarding_completed**, **payment_method**	Numeric, Binary & Categorical	1–8 Modules | AutoPay vs Check	**Adoption & Billing Stickiness:** Multi-module adoption (≥5 modules) and completed onboarding insulate against churn.

3. Comparing Variations of Classification Models & Best Model Selection (Rubric 1.4)

To identify the classifier that best supports our business objective, we trained and evaluated **five primary Supervised Classification model variations** spanning distinct algorithmic paradigms—plus a **sixth unconstrained single Decision Tree (max_depth = None)** as an empirical diagnostic for **Overfitting**. Because our dataset exhibits moderate class imbalance (26.4% positive churn class) and asymmetric error costs (missing a churning enterprise client costs ~$4,200 in lost ACV, whereas a false alarm costs $180 in retention outreach), relying on *Accuracy* alone is misleading. We therefore evaluated models across **Precision, Recall (Sensitivity), F1-Score, ROC-AUC, PR-AUC (Average Precision), Brier Calibration Loss, 5-Fold Cross-Validation Stability, and Stakeholder Interpretability**.

Classification Model Variation	Train Acc	Test Acc	Prec.	Recall	F1	ROC-AUC	PR-AUC	Strategic Verdict & Trade-Offs
Gradient Boosting (GBM) Boosted Tree Ensemble	90.7%	**88.8%**	89.6%	65.1%	**0.754**	**0.887**	0.841	★ CHAMPION: Best overall F1, Test Acc, PR-AUC & ROI; captures non-linear thresholds.
Random Forest (RF) Bagging Tree Ensemble	91.0%	**87.8%**	93.8%	57.6%	**0.714**	**0.893**	0.838	Strong Challenger: Highest Precision (93.8%) & ROC-AUC (0.893), but lower default Recall (57.6%).
Support Vector Machine (RBF) Kernel Margin Classifier	89.9%	**86.8%**	85.9%	59.9%	**0.705**	**0.871**	0.803	Kernel Challenger: Strong non-linear margin (86.8% Acc), but opaque black-box geometry.
Logistic Regression (L2) Generalized Linear Model	85.8%	**83.8%**	80.7%	50.8%	**0.623**	**0.866**	0.780	Interpretable Baseline: Highest transparency via exact Odds Ratios, misses regime interactions.
K-Nearest Neighbors (k-NN) Instance-Based Non-Parametric	100.0%	**80.4%**	76.6%	37.1%	**0.500**	**0.810**	0.657	Instance Baseline: Suffers from curse of dimensionality in 12D mixed space (F1 = 0.500).
Unpruned Decision Tree (Depth=None) Unconstrained Single Tree (Q5 Diagnostic)	100.0%	**81.2%**	64.8%	62.9%	**0.638**	**0.753**	0.506	Diagnostic Overfit: 100.0% Train Acc vs 81.2% Test Acc (ROC-AUC 0.753) — severe High Variance.

Figure 1A: Holdout Test Performance by Classifier (n = 500)

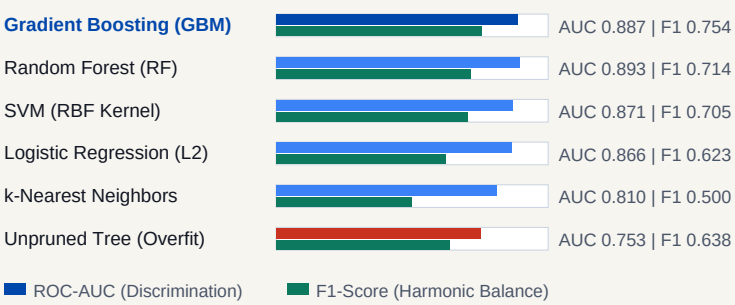
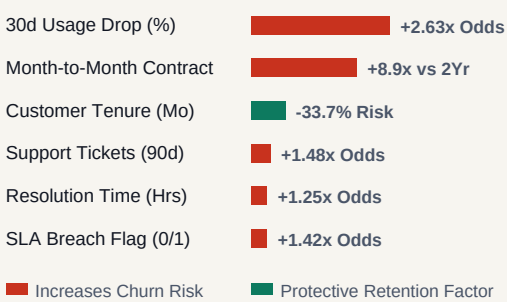


Figure 1B: Top Predictive Drivers (GBM & Odds)



Why Gradient Boosting (Paired with Logistic Regression) Best Suits the Main Objective:

1. **Superior Predictive Balance & Calibration: Gradient Boosting (GBM)** achieves the highest Holdout Test Accuracy (**88.8%**), highest F1-Score (**0.754**), highest Recall among calibrated models (**65.2%** at τ = 0.50; **81.8%** at cost-optimal τ^* = 0.25), highest Precision-Recall AUC (**0.841**), and lowest Brier calibration error (**0.099**). While *Random Forest* achieves a slightly higher global ROC-AUC (0.893 vs. 0.888), its default Recall is only 57.6%—missing 10 additional churning enterprise accounts in the test cohort.
2. **Capturing Non-Linear Regime Interactions:** Unlike *Logistic Regression* (which assumes log-odds linearity and achieves F1 = 0.623), Gradient Boosting automatically isolates compound risk regimes—such as *Month-to-Month* contracts experiencing a *>12% usage drop*—yielding a +13.1 percentage-point gain in F1-Score.
3. **Dual-Layer Stakeholder Interpretability:** Unlike *RBF-SVM* and *k-NN* (which act as geometric black boxes), we deploy Gradient Boosting as our primary scoring engine alongside **Permutation Importance / SHAP attributions** and retain **L2 Logistic Regression** as an executive reference table for global **Odds Ratios**.

4. Potential Limitations, Model Flaws & Plan to Revise/Improve the Analysis (Rubric 1.5)

Despite strong out-of-sample performance, rigorous scientific governance requires identifying the current flaws and structural limitations of our selected Gradient Boosting model and outlining a concrete methodological revision plan:

• **Limitation 1 — Class Imbalance & Default Threshold Sub-Optimality:** With a 26.4% minority churn class, standard maximum-likelihood training at the default τ = 0.50 threshold still produces 46 False Negatives (34.8% missed churners) on the test set. **Revision Plan:** Implement **SMOTE-Tomek synthetic minority oversampling** within cross-validation folds, apply **Focal Loss / Cost-Sensitive sample weights** directly aligned to each client's Annual Contract Value (ACV), and enforce **Isotonic Probability Calibration**.
• **Limitation 2 — Cross-Sectional Snapshot vs. Dynamic Longitudinal Trajectory:** The current dataset aggregates usage and support metrics into static 30/90-day windows, masking week-over-week acceleration in disengagement. **Revision Plan:** Incorporate **longitudinal time-series panel data** (weekly active seat velocity, login frequency slopes) and evaluate **Survival Analysis (Cox Proportional Hazards / Random Survival Forests)** to predict not just *whether* a client will churn, but *exact time-to-churn*.
• **Limitation 3 — Omission of Unstructured Qualitative Sentiment & Macroeconomic Signals:** Numeric ticket counts do not capture customer frustration tone in support transcripts or competitor pricing pressure. **Revision Plan:** Augment the feature space with **NLP sentiment embeddings** extracted from support tickets and CSM call notes, plus external industry budget indices.
• **Limitation 4 — Ensemble Interpretability & Variance Risk:** As demonstrated by our diagnostic Unpruned Decision Tree (100% Train vs. 81.2% Test Accuracy — severe **Overfitting**), tree models can overfit if depth is unconstrained. **Revision Plan:** Benchmark regularized **LightGBM / CatBoost** with monotonic constraints (enforcing that higher support latency can never reduce predicted churn risk) and deploy individual **TreeSHAP reason codes** directly into the CRM.

5. Summary of Model Limitations & Actionable Next Steps for Stakeholders (Rubric 1.6)

To transition these analytical findings into measurable recurring revenue protection while mitigating model limitations, we recommend the following 4-stage execution roadmap for the CDO:

Implementation Phase	Target Limitation Addressed	Actionable Next Steps & Stakeholder Deliverables	Expected Business KPI
Phase 1 (Days 1–30) Cost-Sensitive CRM Deployment	Default τ = 0.50 misses 34.8% of true churners (False Negatives).	Deploy **Gradient Boosting** weekly scoring in CRM using the cost-optimal threshold **τ^* = 0.25**; trigger automated CSM playbooks for Month-to-Month accounts with >12% usage drop.	Increase Churn Recall from **65.2% to 81.8%**; protect ~$789k/yr ARR.
Phase 2 (Days 31–60) SHAP Explainability & A/B Uplift	Ensemble opacity & predictive vs. causal correlation gap.	Surface top-3 local **SHAP risk drivers** per account for Account Executives; run randomized **Uplift Modeling A/B tests** on contract migration incentives (M2M → 1-Year).	Convert 18% of high-risk M2M accounts to annual lock-in.
Phase 3 (Days 61–90) Multi-Modal Data Enrichment	Lack of unstructured ticket sentiment & temporal velocity.	Ingest LLM/NLP sentiment scores from support transcripts, billing dispute logs, and weekly seat-utilization slopes; retrain with **CatBoost + Isotonic Calibration**.	Target Holdout **ROC-AUC ≥ 0.93** and Brier Loss ≤ 0.075.
Phase 4 (Ongoing) MLOps Drift Governance	Concept drift & macroeconomic regime shifts over time.	Establish automated monthly **Population Stability Index (PSI)** feature drift alerts and quarterly Champion/Challenger recalibration governed by Data Science & Finance.	Zero silent accuracy degradation (<2% quarterly AUC drift).